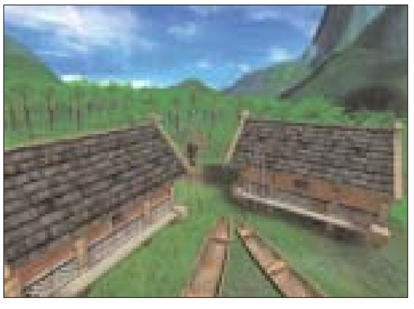
Serious Sam: Second Encounter: The *Serious* engine is also based on the OpenGL API, but is a better engine for evaluating the latest cards. Not only does the engine scale well, it also churns out a lot of high-polygon count monsters, stressing a video card to a great degree. As mentioned earlier, this was the game of choice for evaluating a card's OpenGL performance for the high-end gaming area and was thus given a greater weightage in that sector.
Unreal Tournament 2003 Demo (UT 2003): This is the latest *Unreal* engine that we could get our hands on, since *Unreal 2* wasn't released at the time of testing. Suffice to say, the engine is sufficiently advanced so as to put even the fastest of cards through their paces. Vertex shaders, reflective surfaces, bump-maps, high-polygonal enemies and levels—everything mentioned in the Journal of Greater Eye-candy is implemented with great care by this game. This engine was used to evaluate a card's performance under the DirectX API. This benchmark too, gathered greater weightage for the high-end gaming area.
VulpineGL Mark: This is an OpenGL-based synthetic benchmark. It is used to evaluate the potential performance of a card under advanced features such as hardware texture and lighting, and cube mapping. The test consists of several stressful locales such as a dense forest, reflective water, a high-tech lab with transparencies and lots of light sources and shadows. Such elements strain a video card, giving an indication of its performance while playing an OpenGL-based game. A final score is displayed in the form of an average fps that a card managed to draw.



3DMark 2001 SE: 3DMark 2001SE fills in the boots of a synthetic test for DirectX evaluation. It incorporates the new features that went into the DirectX 8.1 API. This makes it suitable for evaluating the latest graphics cards that claim to be DirectX8.1 and DirectX9.0 compliant, as it runs tests that use a card's hardware T&L unit, its pixel shader unit, various bump mapping techniques and so on. This benchmark skips the relevant test if a particular feature is not supported by the card—a course that is directly reflected in the unified score which is presented at the end of the test.
3D Quality test: To determine the visual quality offered, we tested the cards under *Quake III Arena* and *UT 2003* with anti-aliasing and anisotropic filtering turned on. The features offer more realistic, cleaner pictures. Screenshots from the two games were judged by a panel of three and points were awarded to each card based on the quality of the shots.

To test the graphics cards, we used a test-bed consisting of a P4 2.8 GHz processor sitting on the Intel 850EMV2 motherboard, with 256 MB RDRAM and a 40 GB ATA/100 hard drive, along with a Samsung Samtron 75E 17-inch monitor. The OS was Windows XP Professional with Service Pack 1. The monitor was set at 1024x768 resolution with 75 Hz refresh rate. Vertical Sync was turned off from the respective drivers so that the frame rate while benchmarking, would not get locked to the limit set by the monitor's refresh rate. For testing the nVidia cards, the latest 41.03 Detonator drivers for XP were used. The ATi cards saw the CATALYST 3.0 driver set loaded on the system. Matrox cards were tested with the drivers that came bundled along.